

Dr. Lokendra Chouhan, SMIEEE

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Professional Experience



Assistant Professor, *Indian Institute of Information Technology*, Sri City, AP.

KEY SUBJECTS: Signals and systems, Information Theory and Coding, EMTL, Digital Electronics, Wireless communication systems lab etc. (November 2022-Present)



Postdoctoral Research Fellow, *King Abdullah University of Science and Technology*, Thuwal, KSA, Govt. of KSA.

PROJECT TITLE: Diffusive Molecular Communication and Tera-Hertz Wireless Communications.

MENTOR: Prof. Mohammad Slim Alouini. (July 2021-November 2022)



Senior Research Fellow, *Indian Institute of Technology*, Indore, CSIR Govt. of India.

PROJECT TITLE: Performance Analysis of Diffusive Molecular Communication with Active Receiver.

MENTOR: Prof. Prabhat K Upadhyay. (September 2020 - June 2021)



Teaching Assistant, *Visvesvaraya National Institute of Technology*, Nagpur.

KEY SUBJECTS: Signals and systems, Analog Communication, Digital Electronics, Wireless communication systems lab etc. (July 2017 - September 2020)



PhD Research, *Visvesvaraya National Institute of Technology*, Nagpur.

RESEARCH TOPIC: Performance Analysis and Optimizations in Diffusive Molecular Communication.

MENTOR: Prof. Prabhat K Sharma. (July 2017 - September 2020)



Assistant Professor, *Medi-Caps University*, Indore.

KEY SUBJECTS: Electromagnetic, Communication System, Signals and systems, Analog Communication, Digital Electronics, Wireless communication systems lab etc.

KEY RESPONSIBILITIES: Mentored several B.tech and M.tech students in their final year projects. Involved in supervision of extracurricular activities such as sports and cultural events.

(September 2010- July 2017)



Assistant Professor, *Oriental University*, Indore.

KEY SUBJECTS: Electromagnetic, Control System, VHDL lab etc. (2009-10)



Mentored,

- *Postgraduate thesis*

1. Sunil Kumar (2019-20)

2. Saket Thool (2019-20)

- PhD Ongoing as a Co-Guide-2

Mahesh N., IIIT, Sri City, Naveen, DTU Delhi .

- *Undergraduate project*

1. Himanshu Dhasmana.

2. Shivani (2019-20)

Academic Education

PhD **Visvesvaraya National Institute of Technology**, Nagpur, India.

Doctor of Philosophy, Electronics and Communication Engineering

Specialization: Molecular Communication *Advisor*: Dr. Prabhat K. Sharma

Thesis Title: Diffusive Molecular Communication for Nano-Networks: Performance Analysis and Optimization. (2020)

Masters **Institute of Engineering, DAVV**, Indore, India.

Master of Engineering, Electronics and Communication Engineering

Specialization: Digital Communication *Advisor*: Prof. P D Ashok

Thesis Title: Performance Analysis of Heterogeneous Wireless Networks.

Grade: 76% (2015)

Graduation **Medicaps Institute of Technology and Management (RGTU, Bhopal)**, Indore, India.

Bachelor of Engineering, Electronics and Communication Engineering

Grade: 68.09% (2008)

12th **Govt Higher Secondary School, MP Board**, Satwas, Dewas, India.

Higher Secondary, Physics, Chemistry and Maths.

Grade: 73.4% (2003)

10th **New Diamond Higher Secondary School, MP Board**, Satwas, Dewas, India.

High Secondary,

Grade: 71.2% (2001)

Publications

Journals

- 1 **Lokendra Chouhan** and Mohamed-Slim Alouini, "*Interfacing of Molecular Communication System with Various Communication Systems over Internet of Every Nano Things*", **IEEE Internet of Things Journal**, 2023, DOI: 10.1109/JIOT.2023.3273030 .
- 2 **Lokendra Chouhan**, M. -S. Alouini, "*Rescaled Brownian Motion of Molecules and Devices in Three-Dimensional Multiuser Mobile Molecular Communication Systems*", **IEEE Transactions on Wireless Communications**, vol. 21, no. 12, pp. 10472-10488, Dec. 2022, doi: 10.1109/TWC.2022.3184068.

- 3 **Lokendra Chouhan**, Prabhat Kumar Sharma and Neeraj Varhsney, "*Optimal Transmitted Molecules and Decision Threshold for Drift-Induced Diffusive Molecular Channel With Mobile Nanomachines*", **IEEE Transactions on NanoBioscience**, vol. 18, no. 4, pp. 651-660, doi: 10.1109/TNB.2019.2935241, Oct. 2019 .
- 4 **Lokendra Chouhan**, Prabhat Kumar Sharma, and Neeraj Varhsney, "*On Gradient Descent Optimization in Diffusion-Advection based 3-D Molecular Cooperative Communication*", **IEEE Transactions on NanoBioscienc**, doi: 10.1109/TNB.2020.2996243 .
- 5 **Lokendra Chouhan**, Prabhat Kumar Sharma, Prabhat Kumar Upadhyay, and Parul Garg, "*Impacts of Unintended Nanomachine in Diffusion-based Molecular Communication System*", **IEEE Transactions on Molecular Biology and Multi-scale Communications Systems**, Vol. 6, Issue: 3, Dec. 2020, pp. 210 - 219. doi: 10.1109/TMBMC.2020.3025501
- 6 **Lokendra Chouhan**, Prabhat Kumar Sharma, and Adam Noel, "*Molecular Communication in Fractional Diffusive Channel*", **IEEE Communications Letters**, doi:10.1109/LCOMM.2020.2999577.
- 7 Shivani Dhok, **Lokendra Chouhan**, Adam Noel, and Prabhat Kumar Sharma, "*Co-operative Molecular Communication in Drift-Induced Diffusive Cylindrical Channel*", **IEEE Transactions on Molecular Biology and Multi-scale Communications Systems**, doi: 10.1109/TMBMC.2021.3089939.
- 8 **Lokendra Chouhan**, Prabhat Kumar Sharma, "*Molecular Communication in Three-Dimensional Diffusive Channel with Mobile Nanomachines*", **Elsevier Nano Communication Networks**, Vol. 24, ISSN 1878-7789, doi:https://doi.org/10.1016/j.nancom.2020.100296, March 2020 .
- 9 **Lokendra Chouhan**, Prabhat Kumar Upadhyay, "*Effect of Time-Dependent Drift and Diffusion in Molecular Communication Systems* ", **IEEE Communications Letters**, doi: 10.1109/LCOMM.2022.3142753, January 2022 .
- 10 **Lokendra Chouhan**, Prabhat Kumar Upadhyay, "*Time-Scaled Brownian Motion of Transmitter and Receiver in Anomalous-Diffusive Molecular Communication Channel*", **IEEE Transactions on Molecular Biology and Multi-scale Communications Systems**, vol 8, issue 3, June 2022, pp-207-211, doi: 10.1109/TMBMC.2022.3181506.
- 11 Nadezhda Briantceva, **Lokendra Chouhan**, Matteo Parsani, and Mohamed-Slim Alouini, "*Impact of Evanescence Process on Three-Dimensional Sub-Diffusion based Molecular Communication Channel*", in **IEEE Transactions on NanoBioscience**, March 2023, doi: 10.1109/TNB.2023.3260107 .
- 12 **Lokendra Chouhan**, "*Impact of Anomalous Diffusion Phenomenon on Molecular Information Delivery in Bounded Environment*", in **IEEE Transactions on NanoBioscience**, 2024, doi: 10.1109/TNB.2024.3467695 .
- 13 Preprint: **Lokendra Chouhan**, "*Molecular Communication in Bounded Spherical Region with Anomalous Diffusion Phenomenon*", submitted in **IEEE Transactions on Molecular Biology and Multi-scale Communications**, 2024 .

- 14 Preprint: **Lokendra Chouhan**, “*Localization of Nanomachines in Anomalous Diffusion Based Molecular Communication*”, a major revision submitted in **IEEE Communications Letters**, 2024 .

Conferences

- 1 Nadezhda Briantceva, **Lokendra Chouhan**, Matteo Parsani, and Mohamed-Slim Alouini, “*Mobile Sub-diffusion Molecular Communication Channel*”, In Proceedings of the 10th ACM International Conference on Nanoscale Computing and Communication (NANOCOM '23). Association for Computing Machinery, New York, NY, USA, Article 152-153. <https://dl.acm.org/doi/pdf/10.1145/3576781.3608743>, **Published**.
- 2 Nadezhda Briantceva, **Lokendra Chouhan**, Matteo Parsani, and Mohamed-Slim Alouini, “*Evanescence reaction in subdiffusion molecular communication channel*”, In Proceedings of the 9th ACM International Conference on Nanoscale Computing and Communication (NANOCOM '22). Association for Computing Machinery, New York, NY, USA, Article 28, 1–2. <https://doi.org/10.1145/3558583.3558868>, **Published**.
- 3 **Lokendra Chouhan** and Mohamed-Slim Alouini, , “*On the transport of molecular information in sub-diffusion media with reflecting boundary*”, In Proceedings of the 9th ACM International Conference on Nanoscale Computing and Communication (NANOCOM '22). Association for Computing Machinery, New York, NY, USA, Article 28, 1–2. <https://doi.org/10.1145/3558583.3558868>, **Published**.
- 4 Prabhat Kumar Sharma, **Lokendra Chouhan**, “*Performance Analysis of Diffusive Molecular Communication in Three Dimensional Channel*”, **Asia Pacific Radio Science Conference, URSI, New Delhi, India, March 2019**. doi: <https://www.ursi.org/proceedings/procAP19/papers2019/URSILoke.pdf>, **Published** .

Book Chapter

- 1 Abhinav, **Lokendra Chouhan**, P. Sharma, “*Range Expansion in Neuro-spike Synaptic Communication: Error Performance Analysis*”, Bio-inspired Information and Communications Technologies. BICT 2023, Okinawa, Japan, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol 512. Springer. **Published**.

Research Interests

- 5G/6G wireless communications.
- Semantic Communications
- Molecular communication
- Nano-scale communications/nano-networking
- Internet of bio-nano things (IoBNT) and body-area-networks (BAN) for communications.
- Synaptic and neuro-spike communications.
- Tera-Hertz communications.

Key Coursework

- Communication Systems
- Linear Algebra
- Random Variable Theory
- Digital Logic Design
- Signals and Systems
- Information Theory and Coding
- Electromagnetic Theory
- Circuit Analysis

Professional Services in Research

- IEEE Senior member
Reviewer for
- IEEE Communications Letters
- IEEE Internet of Things Journal
- IEEE Wireless Communications Letters
- IEEE Access
- Elsevier AEU Journal
- Frontier Communications
- IEEE TMBMC
- Served/Serving as a TPC member for ACM Nanocom 2022, Barcelona/ACM Nanocom 2023, Coventry, UK

Professional Activities in Academia

- Chairmen Placement, IIIT Sri City (start date-30th Sept 2024-till date).
- UG Coordinator-cum-SDC Advisor (1 Jan 2024-20th Sept 2024), IIIT Sri City.
- IT Committee member and Co-in charge for IT Procurement at the IIIT Sri City.
- ECE Placement Coordinator, IIIT Sri City.
- Nodal officer, e-Samarth Portal, IIIT Sri City.
- Workshop session titled Internet of Bio Nano-things (IoBNT) organized as workshop coordinator in DCOSS IoT conf. held in TII Abudhabi, UAE, 2024.
- Workshop Conducted: “*Next Generation Wireless and Biological (Wi-Bi) Communications Workshop*” in December 2023 sponsored by SERB and IIIT Sri City, and endorsed by committee of MBMC TC IEEE.
Role: **Coordinator**
Duration: 18-22 Dec. 2023
Place: IIIT, Sri City, India.
- Global Initiative of Academic Networks (GIAN) workshops on “*Fundamentals of Optical Wireless Communication*”.
Role: **Student coordinator**
Duration: 14-19 Jan. 2018
Place: VNIT, Nagpur, India

Technical Expertise

- OS** UNIX, LINUX, WINDOWS (10 years).
- Languages** C (10 years), C++ (8 years).
- Tools** MATLAB (8 years), MATHEMATICA (5 years), L^AT_EX (6 years) etc.
- Platforms** Hands-on experience in NI USRP platform, WiCOMM-T universal communication platform, GSM/GPRS based communication platform, SCIENTECH SDR platform.

Honors and Awards

- October 2023 Elevated to IEEE Senior member.
- July 2020 Nominated from Visvesvaraya National Institute of Technology, Nagpur as a best researcher award for the Ph.D from Ministry of Electronics and Information Society (Meity), Govt. of India.
- July 2017 Received teaching assistant scholarship for Ph.D from Ministry of Electronics and Information Society (Meity), Govt. of India.
- 2013-2015 Successively three times GATE exam qualified from 2013 to 2015.

Work Style

- Well-organized and passionate
- Willing to perform basic tasks and move on to solve complex problems
- Able to learn new knowledge and adapt to new environments quickly
- Strong independent work style and excellent teamwork skills
- Highly committed and strong critical thinking skills

Personal Details

- Address**  S-199, Silicon City, Indore
- Mobile No.**  +91 8889 345 941 **Languages:** English, Hindi
- Email id**  lokendrachouhan22@gmail.com
- Research Gate**  <https://www.researchgate.net/profile/LokendraChouhan3>

References

1. Prof. Mohmed Slim Alouini (Postdoc Mentor)

-  Professor  slim.alouini@kaust.edu.sa
KAUST Thuwal, KSA.  +966 54 470 0038

2. Dr. Prabhat Kumar Sharma (PhD Mentor)

-  Assistant Professor,  prabhatmadhavec1@gmail.com
Department of ECE,  prabhatsharma@ece.vnit.ac.in

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3. Dr. Prabhat K Upadhyay (Research Fellow Mentor and Collaborator)

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Department of EE,
IIT Indore, India.

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Declaration

I hereby declare that the above mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above mentioned particular.

Date: September 27, 2024

Place: Indore (India)

Lokendra Chouhan